Name: Chaitra Samant
Branch: TY Computer Engg
Reg No: 231070055
Batch: B

EXPERIMENT 07 - MIS LAB

AIM: To prepare a comprehensive project plan for an information system: the Customer Complaint Intelligence System (CCIS) for Corporate Decision Making

PROJECT PLAN
Customer Complaint Intelligence System for Corporate Decision Making

THEORY

1. What is a Project Plan?

A project plan is a formal, approved document that defines the project's objectives, scope, schedule, resources, risks, and deliverables. It serves as the primary roadmap for guiding the project team and stakeholders from initiation through closure. A project plan consolidates all planning outputs into a single reference document that enables coordinated execution, proactive risk management, and controlled change. In the context of a Management Information System (MIS), the project plan is especially critical because MIS projects involve cross-functional integration, data governance, AI/ML components, and stakeholder-facing dashboards — all of which require careful orchestration.

The project plan differs from a project schedule (which only covers the timeline) and a project charter (which only covers authorization). A comprehensive project plan encompasses both and extends into detailed planning for feasibility, effort, risk, resources, and technical architecture.

2. Components of a Project Plan

A comprehensive project plan for an information system typically includes the following components:

- Project Overview: A high-level summary of the information system, its purpose, background, and strategic objectives.
- Scope Definition: Clearly delineated boundaries of what is included and excluded from the project, along with the key features to be developed.
- Feasibility Analysis: An assessment of technical, financial, operational, and schedule feasibility to justify project viability before investment.
- Work Breakdown Structure (WBS): A hierarchical decomposition of the total project scope into manageable tasks and work packages.
- Project Schedule (Gantt Chart): A time-phased representation of all tasks, their durations, dependencies, and parallel tracks using a Gantt chart.
- Effort and Cost Estimation (COCOMO): A quantitative model for estimating software development effort (person-months), development time, and team size.
- Risk Analysis: Identification, assessment, and prioritization of risks using the Risk Exposure formula (RE = P x I), along with mitigation strategies.
- Resource Allocation: Assignment of human resources, tools, and infrastructure to project tasks based on skill requirements and availability.

- Tools and Technologies: The technology stack, development platforms, and third-party integrations used to build and deploy the system.
- Deliverables: A structured list of tangible outputs produced at each project phase, from requirements documents to deployment packages.

TASKS

Task 1: Project Overview

The Customer Complaint Intelligence System (CCIS) is a multi-domain, AI-driven Management Information System designed to transform raw customer complaint data into actionable corporate intelligence. Modern enterprises recognize that systematic complaint analysis is a strategic imperative; however, most organizations rely on basic CRM or ticketing tools that merely log issues without uncovering patterns, root causes, or risks.

The CCIS bridges this gap by centralizing complaint data from multiple organizational channels — call centers, web portals, email, CRM systems — and applying AI/NLP-based classification, sentiment analysis, trend detection, churn prediction, and root-cause analysis. Structured insights are then delivered to executives, operations managers, customer service agents, and quality assurance teams through role-based dashboards and automated reports.

The system is developed as a prototype over a 63-day timeline using a modern full-stack architecture: Next.js with Tailwind CSS for the frontend, Supabase (managed PostgreSQL) for the database and authentication, Python with LangChain for AI/NLP pipelines, and Vercel for deployment. The project team consists of three members working across three two-week agile sprints.

Task 2: Scope

In-Scope

- Consolidation of complaints from all internal channels (call center, email, web portal, CRM) into the centralized CCIS database.
- Automated analytics modules: AI/NLP complaint categorization, sentiment analysis, recurring issue detection, root-cause analysis, SLA monitoring, and churn risk prediction.
- Executive dashboards and KPI reports providing summarized, decision-ready insights for management.
- Integration APIs and ETL connectors with existing corporate CRM, ticketing systems, and ERP for bi-directional data exchange.
- Role-based user interfaces for customer service agents, operations managers, data analysts, and executives.
- Feedback loop capturing resolved root causes and policy outcomes for continuous improvement.
- Security controls including role-based access control (RBAC), audit logging, and data anonymization for PII.

Out-of-Scope

- Real-time chatbots or live customer support tools (the system analyzes logged complaints, not live conversations).
- Social media or third-party complaint aggregation (initial version focuses on internal enterprise channels).
- Direct customer service actions such as issuing refunds or dispatching replacements.
- Replacement of existing CRM systems; the CCIS integrates with rather than replaces them.
- Hardware procurement beyond the infrastructure required to host the software (cloud/Vercel/Supabase).

Task 3: Feasibility Analysis

Technical Feasibility

The CCIS employs a proven, mature technology stack. Next.js and Tailwind CSS are industry-standard tools for building responsive, performant web applications. Supabase provides a managed PostgreSQL database with built-in authentication, REST APIs, and row-level security, eliminating the need for manual server management. Python and LangChain are the de facto standards for LLM-based NLP workflows. All tools offer free tiers suitable for the prototype phase, and the architecture is horizontally scalable for production deployment. The team has the required competencies across full-stack development, AI integration, and database design.

Financial Feasibility

The prototype can be built and deployed at near-zero infrastructure cost. Vercel's Hobby plan, Supabase's free tier, and GitHub Actions free minutes cover all hosting, CI/CD, and database needs. LLM access via Hugging Face (open models) or OpenAI trial credits requires no upfront licensing cost. The primary cost is human effort: a 3-person team over approximately 9.72 months of cumulative person-months (COCOMO estimate detailed in Task 6). For an organizational deployment, cloud costs would scale proportionally, and the financial return from reduced customer churn, faster complaint resolution, and data-driven decision savings significantly justifies the investment.

Operational Feasibility

The CCIS is designed to minimize workflow disruption. Customer service agents continue logging complaints through familiar interfaces; the system adds analytical value without replacing operational processes. Role-based dashboards present insights tailored to each user level, reducing cognitive overhead. A phased

training program and intuitive UI design address the user adoption risk (R03) identified in the risk analysis. Senior management and operations managers gain direct productivity benefits through decision-ready dashboards, creating organizational buy-in.

Schedule Feasibility

The project is organized into three two-week agile sprints over a 5-6 week prototype timeline, with a critical path duration of 63 days. Key parallel development tracks (T5 Frontend, T6 AI/NLP Engine, T7 Dashboard running simultaneously in Weeks 4-7) are used to compress total calendar time. The COCOMO-estimated development time of approximately 9.72 months reflects the full-scale organizational system; the prototype scope is deliberately narrowed to the core features to remain achievable within the student timeline.

Task 4: Work Breakdown Structure (WBS)

The WBS decomposes the entire CCIS project into 12 manageable tasks across five phases: Planning, Design, Development, Testing, and Closure. This WBS is consistent with the scheduling performed in Experiment 06.

Task	Task Name	Days	Phase	Depends On
T1	Requirements Gathering & Stakeholder Analysis	6	Planning	—
T2	Feasibility Study & System Scope Definition	4	Planning	T1
T3	System Architecture & Data Flow Design	7	Design	T2
T4	Complaint Database Schema Design & Setup	5	Design	T3
T5	Complaint Intake Portal (Frontend UI)	10	Development	T3
T6	AI/NLP Complaint Classification Engine	14	Development	T4
T7	Analytics Dashboard & Reporting Module	10	Development	T4
T8	Integration of All Modules	6	Development	T5, T6, T7
T9	Unit & Module Testing	5	Testing	T8
T10	System Testing, UAT & Stakeholder Review	7	Testing	T9
T11	Bug Fixing, Optimization & Security Audit	5	Testing	T10
T12	Deployment, Training & Documentation	4	Closure	T11

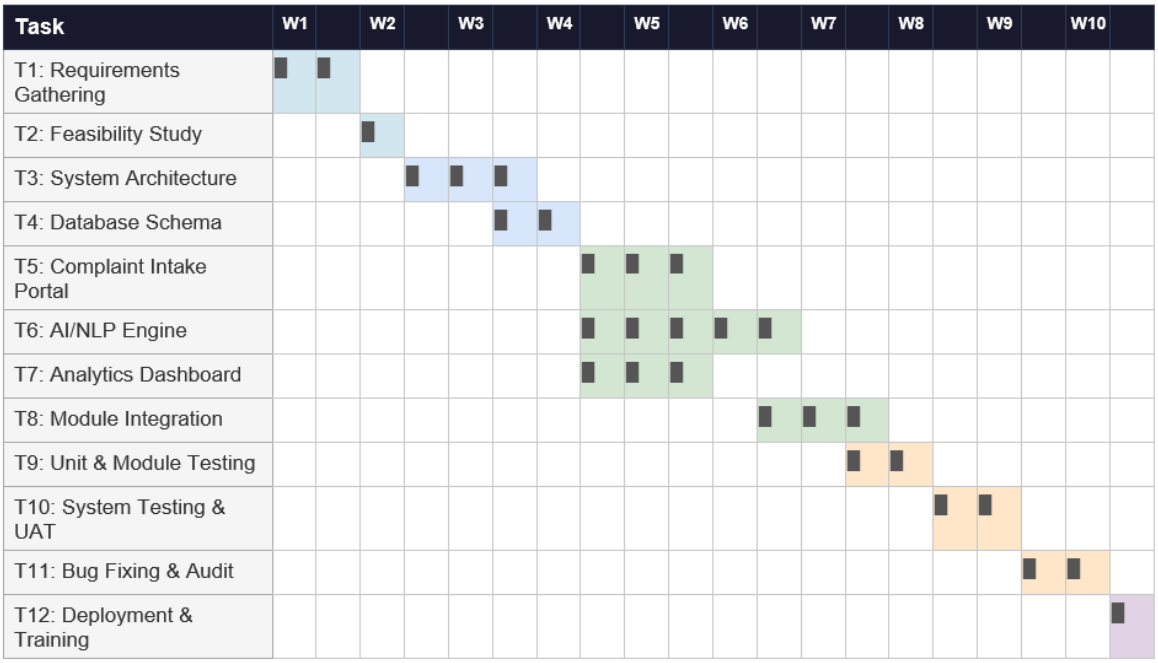
Total Project Duration (Critical Path):

T1 > T2 > T3 > T4 > T6 > T8 > T9 > T10 > T11 > T12 = 63 days

Note: T5 (Complaint Intake Portal), T6 (AI/NLP Engine), and T7 (Analytics Dashboard) run in parallel after T4, substantially compressing the schedule. T5 and T7 each have a float of 6 days.

Task 5: Project Schedule (Gantt Chart)

The Gantt chart below represents the full 63-day project schedule. Shaded blocks indicate that the task is active in that time window. T5, T6, and T7 run in parallel (Weeks 4-7); T8 Integration begins only after all three complete.



Legend: Planning (blue) | Design (light blue) | Development (green) | Testing (orange) | Closure (purple)

Task 6: Effort and Cost Estimation (COCOMO)

The COCOMO (Constructive Cost Model) Basic Model is applied to estimate effort, development time, and team size for the CCIS. The system is classified as Semi-detached because it integrates both standard CRUD operations (complaint registration, user management) and technically complex AI/NLP components (LangChain pipelines, sentiment analysis, churn prediction), placing it between the organic and embedded project types.

COCOMO Formulae (Semi-detached)

- Effort: E = 3.0 x (KLOC)^1.12 [Person-Months]
- Development Time: TDev = 2.5 x (E)^0.35 [Months]
- Team Size: N = E / TDev [People]
- Productivity: P = KLOC / E [KLOC per Person-Month]

COCOMO Estimation Table

Parameter	Value	Explanation
Project Type	Semi-detached	System integrates AI/NLP, web frontend, and a backend analytics engine — moderate novelty and complexity
KLOC Estimate	8 KLOC	~8,000 lines of code covering frontend, APIs, AI pipelines, dashboard, and database logic
Effort (PM)	a = 3.0, b = 1.12	E = 3.0 x (8)^1.12 = 3.0 x 10.28 = 30.84 person-months
Development Time (TDev)	c = 2.5, d = 0.35	TDev = 2.5 x (30.84)^0.35 = 2.5 x 3.89 = 9.72 months
Team Size	N = E / TDev	N = 30.84 / 9.72 = 3.17 => approx. 3 people

Productivity	KLOC / PM	8 / 30.84 = 0.26 KLOC per person-month
Actual Timeline	~9 weeks	63-day critical path (~9 working weeks) — consistent with TDev estimate

Cost Estimation

Assuming an average developer cost of INR 10,000 per person-month (entry-level to mid-level in India):

- Total Estimated Effort: 30.84 person-months
- Total Estimated Cost: 30.84 x 10,000 = INR 3,08,400 (approx.)
- Prototype Cost (3 members, ~3 person-months): INR 50,000 (approx.) — primarily opportunity cost; actual infrastructure cost is near-zero using free tiers.

The 63-day critical path (approximately 9 working weeks) aligns well with the COCOMO TDev estimate of 9.72 months for the full-scale system, confirming that the prototype represents a proportionally scoped subset of the full organizational deployment.

Task 7: Risk Analysis

The following risk assessment is carried forward from Experiment 05, using the Risk Exposure formula RE = Probability (P) x Impact (I), where High = 3, Medium = 2, Low = 1. Risks are listed in priority order from Critical to Medium.

ID	Category	Risk Description	Prob.	Impact	RE	Treatment
R01	Data Quality	Incomplete/inconsistent data logging across service channels	High	High	Critical (9)	Mitigate
R02	Technical/AI	AI algorithms misinterpreting nuanced sentiment or sarcasm	Medium	High	High (6)	Mitigate
R03	Operational	Employees resisting new analytical workflows	Medium	High	High (6)	Mitigate
R05	InfoSec	Privacy concerns and data breaches from consolidated database	Low	High	High (3)	Avoid
R06	Compliance	Failure to meet GDPR/DPDP Act regulations	Low	High	High (3)	Avoid
R04	Infrastructure	Technical difficulties integrating legacy CRM systems	Medium	Medium	Medium (4)	Mitigate
R07	Strategic	Management over-relying on automated AI insights	Low	High	Medium (2)	Mitigate
R08	Reputational	Failure to act on systemic root causes uncovered by CCIS	Low	High	Medium (2)	Mitigate

Key Risk Mitigation Highlights

- R01 (Critical): Enforce strict input standardization and mandatory data validation at every CRM integration point.
- R02 (High): Implement a human-in-the-loop model for anomalous complaints; continuously retrain NLP models with updated datasets.
- R03 (High): Engage end-users early in design; provide role-based training and demonstrate system value through intuitive dashboards.
- R05 (High - Avoid): Enforce RBAC, conduct regular penetration tests, and mandate automated PII anonymization before data reaches the analytics engine.
- R06 (High - Avoid): Embed GDPR/DPDP Act compliance into system architecture from inception — immutable audit logs, MFA, and data retention policies.
- R04/R07/R08 (Medium): Allocate dedicated integration phases; combine AI insights with mandatory expert review; establish a governance framework for acting on system-uncovered root causes.

Task 8: Resource Allocation

Resources are allocated across all three sprints based on the skill requirements of each task. The team of three core members is supplemented by a part-time QA engineer and data analyst during later sprints.

Role	Count	Duration	Responsibilities
Project Manager	1	End-to-End	Planning, scheduling, stakeholder communication, risk oversight
Full-Stack Developer	1	Weeks 3–8	Next.js frontend, Supabase integration, API development
AI/NLP Engineer	1	Weeks 3–7	LangChain pipelines, sentiment analysis, complaint classification
QA Engineer	1	Weeks 7–9	Unit testing, UAT coordination, bug tracking
Data Analyst	1	Weeks 4–8	Dashboard design, KPI definition, reporting module

Infrastructure Resources

- Frontend Hosting: Vercel Hobby Plan (free) — Next.js SSR/SSG with built-in CI/CD.
- Database & Auth: Supabase free tier — managed PostgreSQL, Row-Level Security, REST APIs.
- AI/LLM Access: Hugging Face Hub (free open models) and/or OpenAI trial credits via LangChain.
- Version Control & CI: GitHub (free) with GitHub Actions (2,000 min/month free).
- Project Management: GitHub Issues / Trello for sprint tracking and task management.

Task 9: Tools and Technologies

Frontend

- Next.js 14 (React framework) — server-side rendering, static generation, API routes.
- Tailwind CSS — utility-first CSS for rapid, responsive UI development.

Backend & Database

- Supabase (managed PostgreSQL) — complaint storage, user authentication, Row-Level Security, instant REST APIs.
- Python (serverless functions / Edge Functions) — business logic, data transformation, AI pipeline orchestration.

AI & NLP

- LangChain (Python) — unified framework for orchestrating LLM calls, prompt chaining, and model switching.
- Hugging Face Hub — open-source NLP models for complaint classification and sentiment analysis (zero cost).
- OpenAI API (optional) — GPT-3.5/4 access for advanced summarization and topic detection via LangChain.

DevOps & Deployment

- Vercel — zero-configuration deployment platform for Next.js with built-in CI/CD.
- GitHub Actions — automated pipelines for linting, testing, and deployment on every commit.
- Docker (optional) — containerization for reproducible local development environments.

Testing

- Jest — unit and integration testing for React/Next.js components.
- PyTest — unit testing for Python backend logic and AI pipeline functions.

Project Management

- GitHub Issues & Milestones — sprint tracking, bug reporting, and feature planning.
- Figma — UI/UX wireframing and dashboard prototype design.

Task 10: Deliverables

The following deliverables are produced across the five project phases, consistent with the scoping output of Experiment 02.

ID	Deliverable	Description	Depends On
D1	Requirements Specification	Functional & non-functional requirements document	T2
D2	System Architecture Diagrams	Data flow, general & functional perspective diagrams	T3
D3	Complaint Database Schema	Postgres schema via Supabase, ER diagram	T4
D4	Working Web Application	Next.js + Tailwind frontend, Supabase-backed APIs	T5, T8

D5	AI/NLP Classification Engine	LangChain-based complaint classifier and sentiment analyser	T6
D6	Analytics Dashboard	Executive KPI dashboard and reporting module	T7
D7	Test Reports	Unit, integration, and UAT reports	T9, T10
D8	Security Audit Report	Penetration test results, RBAC configuration	T11
D9	Deployment Package	Vercel deployment, CI/CD pipelines, GitHub Actions config	T12
D10	User & Maintenance Docs	Training materials, API documentation, maintenance guide	T12

CONCLUSION

Hence, in this experiment, the successful development of the Customer Complaint Intelligence System (CCIS) was driven by the integration of four essential planning types: **Feasibility, Structural, Quantitative, and Risk-based analysis.**

By conducting a four-dimensional feasibility study, the project confirmed that the technical stack and financial constraints were aligned with operational goals. The Work Breakdown Structure (WBS) then organized the project into five distinct phases, enabling parallel development to compress the 63-day schedule.

This was reinforced by COCOMO quantitative estimation, which validated the person-month effort and team size requirements, ensuring the timeline remained realistic. Finally, a structured risk analysis categorized potential threats to establish proactive mitigation strategies. Together, these planning types transformed a complex AI concept into a cohesive, actionable document, proving that rigorous preparation is the catalyst for turning raw complaint data into a strategic corporate asset.